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Influence-Aware RAG: A Diagnostic Framework for Document Contribution Analysis

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1. Introduction

Retrieval-Augmented Generation has emerged as a practical solution to language model hallucination by grounding model outputs in external documents. Typical deployments span legal analysis, medical diagnostics, financial analytics and enterprise knowledge systems.

However, as RAG pipelines grow more complex with rerankers, hybrid retrieval, and longer contexts, understanding which documents actually shape generated answers becomes increasingly untraceable to the source material, limiting diagnostic capability and trustworthiness.

2. Literature Survey

RAG mitigates LLM hallucinations by grounding generation in external databases, driving a sharp rise in research activity, with over 1,200 papers published in 2024 alone. Recent work increasingly favours modular architectures: AU-RAG enables dynamic knowledge selection, while MedGraphRAG reports an 8% performance gain through graph-based retrieval mechanisms.

Within these pipelines, reranking plays a central role in refining document order, and **ablation** studies show that removing context evaluators measurably degrades original context grounding, however, retrieval quality metrics correlate weakly with a document's actual contribution during generation, leading to questions about underlying RAG performance.

Overall, current research prioritises architectural improvements over diagnostics. While systems like M-RAG coordinate multi-agent retrieval, **there are no existing frameworks that quantify the divergence between retrieval rankings and true generation-time influence.**

2. Literature Survey (Continued)

Gaps Identified

Attribution-Retrieval Divergence

Current frameworks do not quantify divergence between retrieval rankings and **generation-time document influence** across pipeline configurations. Spearman correlations between retrieval scores and ablation-based influence remain unmeasured.

Diagnostic Versus Architectural Focus

Existing RAG research (1, 2) prioritizes building better RAG systems rather than diagnosing existing ones. Ablation-based attribution provides non-redundant diagnostic information beyond RAG metadata, yet no systematic framework makes this actionable for practitioners.

Evaluation Methodology

Current RAG evaluation literature reviews (3, 4) do not account nor test for document-level influence. Ablation-based attribution scores, dominance ratios, and influence entropy lack benchmark comparisons across datasets and models.

1. **ArXiv 2507.04480 (2025). Source Attribution in Retrieval-Augmented Generation.**
2. **Wu, J., Zhu, J., & Qi, Y. (2024). Medical Graph RAG: Towards Safe Medical Large Language Model via Graph Retrieval-Augmented Generation.**
3. **Frangou, E., et al. (2025). A Systematic Literature Review of Retrieval-Augmented Generation: Techniques, Metrics, and Challenges. *Information* 9(12):320. *MDPI*. DOI: 10.3390/info9120320. [December 12, 2025]**
4. **Preprints.org 202512.0359 (2025). Retrieval Augmented Generation (RAG) and Large Language Models (LLMs) for Enterprise Knowledge Management and Document Automation: A Systematic Literature Review.**

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3. Objectives

Primary Objective

To design and implement a lightweight ablation-based diagnostic tool that measures the rank-level agreement between document retrieval scores and generation-time document influence in a standard **Retrieval-Augmented Generation (RAG) pipeline**, using controlled experiments on a fixed dataset and model.

3. Objectives (Continued)

1. Develop Ablation-Based Influence Metrics

Design and implement reproducible metrics that capture document contribution patterns invisible to standard RAG evaluation.

Core Metrics:

- **Influence score:** Magnitude of output change when a document is removed from context
- **Document influence variance:** Distribution spread across retrieved documents — measuring contribution balance

2. Quantify Attribution-Retrieval Divergence

Measure the correlation between retrieval/reranking scores and actual document influence on generated outputs.

Core Metrics:

- **Spearman correlation** between retrieval scores and ablation-based influence scores

- Quantitative evidence that retrieval metadata alone is **insufficient** for predicting generation-time contribution

4. Methodology

1. Setup
- Choose dataset (Natural Questions)
- Select 100-200 test queries
- Design a standard baseline RAG pipeline using BM25
- Choose a cheap LLM (GPT-OSS 20b, Llama 3.2 8b)

2. For Each Query:

A. Retrieval Phase
- Run retrieval → get top-10 documents
- Record retrieval scores for each document
- Rank documents by retrieval score

B. Baseline Generation
- Feed ALL documents + query to LLM
- Generate baseline answer
- Save this as reference

C. Ablation Loop
- For each document in corpus (D1, D2, ... D10):
  - Remove that document from context
  - Generate new answer
  - Compare to baseline using **ROUGE-L**
  - Difference = influence score for that document

D. Rank Influence
- Rank documents by influence score
- Compare retrieval ranks vs. influence ranks

3. Calculate Metrics
- **Spearman correlation** between retrieval ranks and influence ranks
- Dominance ratio = **max(influence) / sum(influence)**
- Influence entropy = **distribution spread**

4. Detect Divergence
- Flag queries where $\rho < 0.7$
- Identify failure modes (high retrieval, low influence)

5. Aggregate Results
- Count % queries with $\rho < 0.7$
- Compare distributions across configurations
- Generate visualizations and case studies

6. Statistical Analysis
- Test if configuration differences are significant ($p < 0.05$)
- Document failure mode frequency
- Write findings

5. Project Novelty

Existing RAG projects build better retrievers and rerankers. This project diagnoses existing systems by examining retrieval scores versus actual document influence during generation with an **ablation-based framework** across different model and RAG evaluation configurations.

6. Sustainable Development Goals

SDG 9: Industry, Innovation, and Infrastructure

Improved RAG diagnostics support reliable AI-assisted decision-making for urban planning and governance systems (infrastructure analysis, regulatory compliance, public service documentation)

Provides open-source framework enabling researchers and practitioners in developing regions to validate RAG systems without expensive proprietary tools

SDG 11: Sustainable Cities and Communities

Enables transparent evidence-based retrieval for policy documents, ensuring AI tools ground recommendations in verifiable sources

Thank you